

WONJU, Korea, Republic of

WMO: 471140

Lat: 37.3375N

Lon: 127.9467E

Elev: 492

StdP: 14.44

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	11.6	-8.2	3.6	18.0	-4.0	4.5	19.6	15.7	34.1	13.1	29.6	2.3	50	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	91.4	76.2	88.8	74.5	86.4	73.3	78.9	86.0	77.6	84.4	76.4	82.8	4.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	143.8	81.9	75.6	136.3	80.8	74.6	131.8	80.2	43.0	85.6	41.5	83.9	40.3	83.5	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.7	11.0	9.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			2.7	94.8	4.4	2.6	-0.5	96.7	-3.1	98.2	-5.5	99.7	-8.7	101.6
			1.6	80.3	4.2	2.4	-1.4	82.0	-3.9	83.4	-6.2	84.7	-9.3	86.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	26.6	31.8	41.9	54.0	64.2	72.4	77.8	78.5	69.0	57.1	43.4	30.3
	DBStd	19.08	7.38	7.24	6.72	6.62	4.85	4.02	3.95	4.61	5.04	6.31	7.67	7.43
	HDD50	2387	727	511	263	34	0	0	0	0	17	225	610	
	HDD65	5243	1192	930	715	334	73	4	0	0	20	251	649	1075
	CDD50	3860	0	1	13	154	442	673	863	882	570	237	25	0
	CDD65	1241	0	0	0	3	50	226	398	418	140	6	0	0
	CDH74	10043	0	0	0	75	503	1650	3274	3713	795	33	0	0
	CDH80	3111	0	0	0	7	80	444	1065	1368	147	0	0	0
(14) Wind	WSAvg	3.1	2.9	3.1	4.0	4.1	3.6	3.0	3.1	2.9	2.6	2.5	2.6	2.5
(15) Precipitation	PrecAvg	50.4	0.9	1.1	2.0	3.4	3.7	5.6	13.1	10.0	5.9	1.8	1.7	1.2
	PrecMax	81.4	3.0	3.3	4.4	8.3	8.9	16.9	24.2	28.3	23.0	6.5	3.6	3.1
	PrecMin	35.4	0.2	0.0	0.6	0.5	0.8	1.1	3.1	1.5	0.2	0.1	0.3	0.2
	PrecStd	9.9	0.8	0.9	1.1	2.1	2.0	4.0	5.9	6.0	5.3	1.4	1.0	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	57.5	68.4	80.6	85.7	89.7	94.8	95.0	87.4	77.4	67.7	53.5
		MCWB	41.5	48.2	52.2	59.6	63.3	68.5	77.5	76.6	72.6	62.8	56.1	45.4
	2%	DB	43.6	52.0	63.3	75.8	82.2	86.9	91.0	92.1	84.0	74.1	62.8	48.5
		MCWB	37.0	41.7	48.8	57.2	61.8	68.2	76.6	76.6	70.2	61.4	53.4	42.0
	5%	DB	40.6	47.7	59.0	71.9	79.4	84.7	88.1	89.8	81.1	71.3	59.6	44.9
		MCWB	35.0	39.1	46.5	54.7	60.9	67.7	75.0	75.9	68.3	59.8	51.3	39.6
	10%	DB	37.6	43.5	55.3	68.2	76.7	82.5	85.8	87.3	78.7	68.2	56.2	41.7
		MCWB	32.8	36.2	44.3	52.8	59.9	67.1	74.5	75.4	67.4	58.5	49.4	37.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	51.4	55.4	65.3	69.9	75.3	80.9	81.0	76.9	67.4	58.9	47.1
		MCDB	46.0	54.6	62.1	74.6	77.3	80.9	88.5	88.1	81.7	71.6	63.5	51.4
	2%	WB	38.5	43.1	51.6	60.5	66.4	73.2	78.9	79.2	74.1	64.2	55.6	43.2
		MCDB	42.2	50.3	59.1	69.7	75.3	79.8	85.8	86.7	80.0	70.3	60.8	47.0
	5%	WB	35.8	39.6	48.2	57.5	64.3	71.6	77.6	78.0	71.8	61.9	52.8	40.2
		MCDB	39.4	46.1	56.7	67.5	73.1	78.8	84.1	85.1	77.4	68.7	57.7	44.4
	10%	WB	33.4	36.9	45.3	55.0	62.5	70.2	76.3	76.9	69.7	59.8	50.3	37.2
		MCDB	37.0	42.8	53.7	64.7	72.1	77.9	82.5	83.8	76.0	66.2	55.2	41.1
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	19.0	20.1	22.8	21.4	17.9	13.5	14.2	17.1	20.5	18.7	16.8
		MCDBR	18.3	23.9	26.4	28.9	26.5	22.3	17.6	17.8	19.9	23.1	22.9	20.1
	5% WB	MCWBR	12.8	15.1	14.8	13.5	10.4	8.1	6.7	6.3	8.1	11.4	14.0	14.3
		MCWBR	16.3	21.0	23.0	24.0	20.9	16.9	14.9	15.3	15.7	19.0	19.3	19.0
(40) Clear-Sky Solar Irradiance	taub	0.409	0.463	0.552	0.589	0.597	0.658	0.625	0.554	0.478	0.465	0.443	0.404	
		1.995	1.896	1.742	1.712	1.735	1.668	1.800	1.982	2.130	2.033	2.022	2.039	
	Ebn at Noon	232	234	225	227	228	214	220	233	245	232	219	223	
		42	53	68	74	73	78	68	56	45	45	41	38	
(44) All-Sky Solar Radiation	RadAvg	793	1033	1322	1548	1747	1623	1281	1326	1251	1115	782	706	
	RadStd	55	89	140	121	154	136	186	188	150	83	89	55	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.12	N/A	-3.11	N/A	-1.39	-1.82	N/A	-220	+216	+121
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+88

Nomenclature: See separate page